
LONG-RANGE ROBUST UNDERWATER COMMUNICATION

D4 – Final demonstration

Version 1.2 (3 January 2022)

Mandar Chitre, Gabriel Chua, Venugopalan Pallayil

ARL, National University of Singapore



ARL Project Report

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CHAPTER 1

INTRODUCTION

The primary aim of the long-range robust underwater communication project is to develop and test technologies that may be used to communicate with underwater assets robustly, and at potentially long ranges. While we explored technologies such as spatial diversity, link adaptation and disruption tolerance to aid in this, the key focus was on the development of a low-frequency long-range communication system. In deliverable D3, we outlined the design of such a system in detail, and laid out a plan for testing the system in Singapore waters. In Chapter 2 of this report, we detail findings from the final experiment where we tested and demonstrated such a system successfully.

The ability to predict communication performance for a given environment is very useful for operational planning. During the course of this project, we developed a communication performance prediction tool. In Chapter 3, we also compare the prediction from this tool with the performance we achieved during the long-range communication experiments.

Finally, in Chapter 4, we summarize the key findings from the project, and make recommendations on how the techniques developed in this project may be used in other projects and in operational systems.

Do note that in the original schedule, deliverable D4 was also to report on the experimental work on Disruption Tolerant Networks (DTN). Due to an extension to the project needed for the long-range communications work, the DTN experiment was completed in time for deliverable D3, and so the report was included in D3.



Figure 1.1: Low-frequency modem sitting on the deck of the boat on recovery after a long day of acoustic transmissions over 10–20 km range underwater.

CHAPTER 2

FINAL DEMONSTRATION

2.1 Experimental setup

In September 2021, we conducted the final experiment for the project to test and demonstrate a long-range communication system operating in Singapore waters at ranges up to about 17.6 km. The experiment was divided into two phases:

1. (6th, 8th and 10th September)

- Communication performance measurement of the two proposed low-frequency FH-BFSK schemes from deliverable D3 at various ranges.
- Communication performance measurement of the default FH-BFSK and OFDM schemes supported by Subnero modems at various ranges.
- Various environmental and acoustic measurements to aid with performance modeling and interpretation of the experimental results.

2. (27th and 29th September)

- Demonstration¹ of the selected FH-BFSK scheme for reliable simplex data transfer using erasure correction codes, at various ranges.

¹Various colleagues from DSO and DSTA were present onboard RV Galaxea to witness the demonstration on these two days.

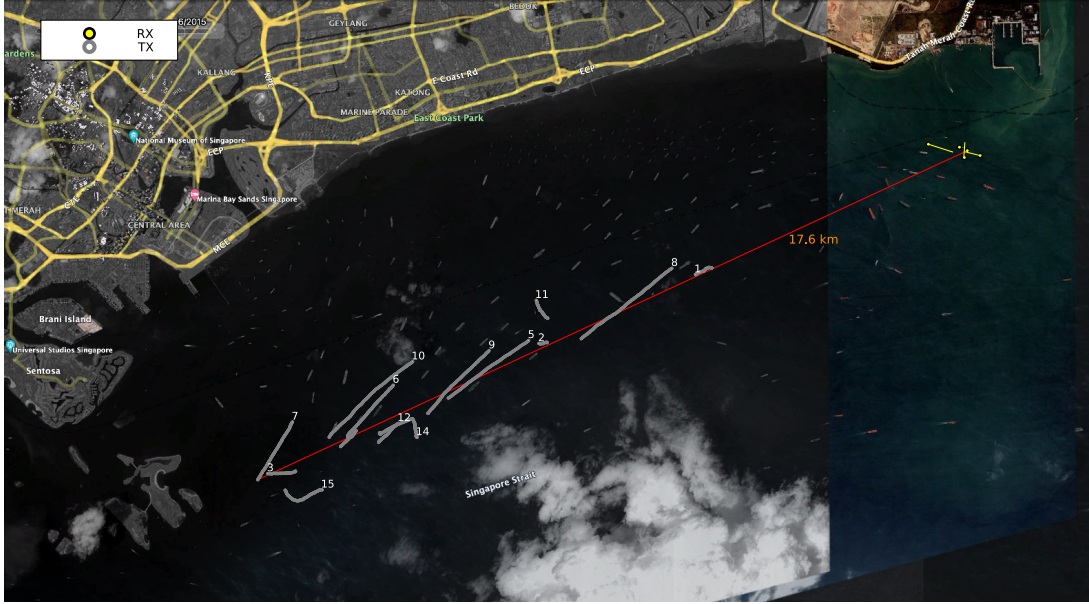


Figure 2.1: Overview of various datasets collected at different ranges during the 5-day experiment. The receiver “RX” location is shown in yellow at the eastern end of the operational area. Due to slightly different anchoring locations on different days, and strong winds and currents leading to some motion, we can see a small amount of variability in the receiver position. The transmitter “TX” was drifting during the transmissions, and its location is seen as grey tracks at various ranges from the receiver. Each track is labeled with a dataset number near its terminal location. All tracks were approximately along a straight line that denotes the operational area, with a maximum communication range of about 17.6 km, from near Kusu island (western limit) to Changi (eastern limit).

On all five days, we used RV Galaxea and RV Kenki 28 to deploy Subnero² low-frequency modems for the demonstration. Due to prevailing COVID-19 related restrictions on boarding RV Kenki 28, we opted to have the receiver system, deployed from an anchored RV Kenki 28, be completely unmanned and autonomous. The receiver system was battery operated, and simply recorded all **real-time communication results** in a log file on a SD card for inspection upon recovery. In addition, the system also recorded acoustic data from 3 external hydrophones continuously, permitting detailed acoustic analysis of specific data sections to aid understanding and interpretation of experimental results. The transmitter system was deployed from RV Galaxea and manually controlled to make transmissions from various ranges. A GPS logger onboard RV Galaxea tracked its location and allowed us to compute ranges between transmitter and receiver at all times during the experiment.

As shown in Figure 2.1, the receiver was deployed from an anchored RV Kenki 28 at the eastern edge of the operational area, and the transmitter was deployed from RV Galaxea, drifting while transmitting from various ranges of up to 17.6 km from the receiver. Several datasets (about an hour each) were collected over the five experiment days. Sound speed profiles were collected at several locations using a CTD probe. An AIS receiver positioned at St Johns Island recorded ship positions in the area during the

²We would like to acknowledge colleagues from Subnero for their support and help in testing the proposed low-frequency FH-BFSK schemes onboard modems provided by them, and for their assistance at sea during the field experiment.

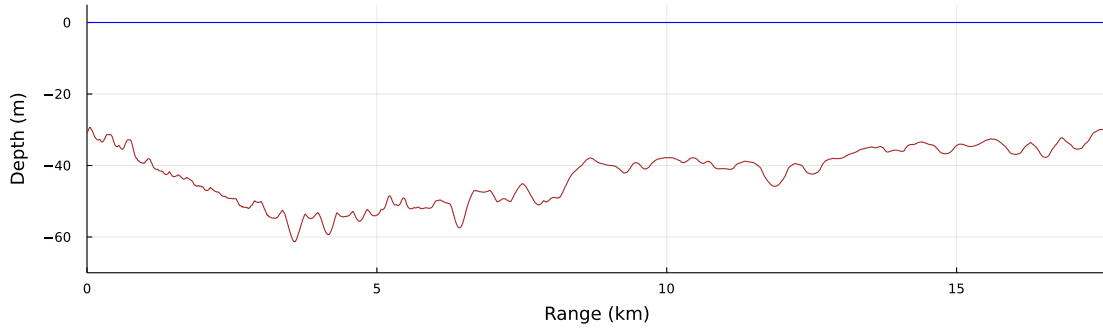


Figure 2.2: Water depth as a function of range along the operational area (along the red line shown in Figure 2.1), with the left side (0 km) being the western end near Kusu Island, and the right side (17.6 km) being the eastern end near Changi.

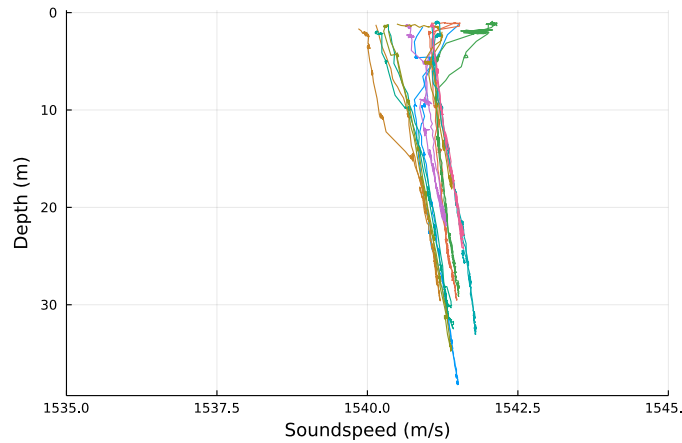


Figure 2.3: CTD measurements at different locations and on different days show an almost iso-velocity sound speed profile, with a very marginal increase in sound speed with depth, and a little variability near the surface. For all practical purposes, the profile can be considered iso-velocity.

experiment. Soundings from MPA charts provided bathymetric data (see Figure 2.2), along with depth sounder readings from RV Galaxea. The weather was mostly fair, with plenty of breeze and occasional rain, but no significant white-capping. The currents were mild (about 0.5 kts) during the first phase (3 days) of the experiment, and stronger (about 1-1.5 knots) during the second phase (2 days). As shown in Figure 2.3, all CTD measurements showed an almost iso-velocity sound speed profile, with a nominal sound speed of 1541 m/s.

The receiver system consisted of a Subnero low-frequency modem with a battery pack (expected battery life >6 hours) and 3 external hydrophones (in addition to the primary transducer of the modem). The hydrophones and the modem formed a linear vertical array, with hydrophone #1 mounted about 4 m above the modem transducer, hydrophone #2 mounted about 5 m below the modem transducer, and hydrophone #3 mounted about 7 m above the modem transducer. The entire array was deployed with the hydrophone #3 (top hydrophone) at about 9 m depth during phase 1 (days 1–3), and 12–15 m (depending on prevailing currents) depth during phase 2 (days 4–5). The transmitter system consisted of a Subnero low-frequency modem (4.5–5.5 kHz band, 185 dB re 1 μ Pa @ 1 m nominal source level) powered using surface power

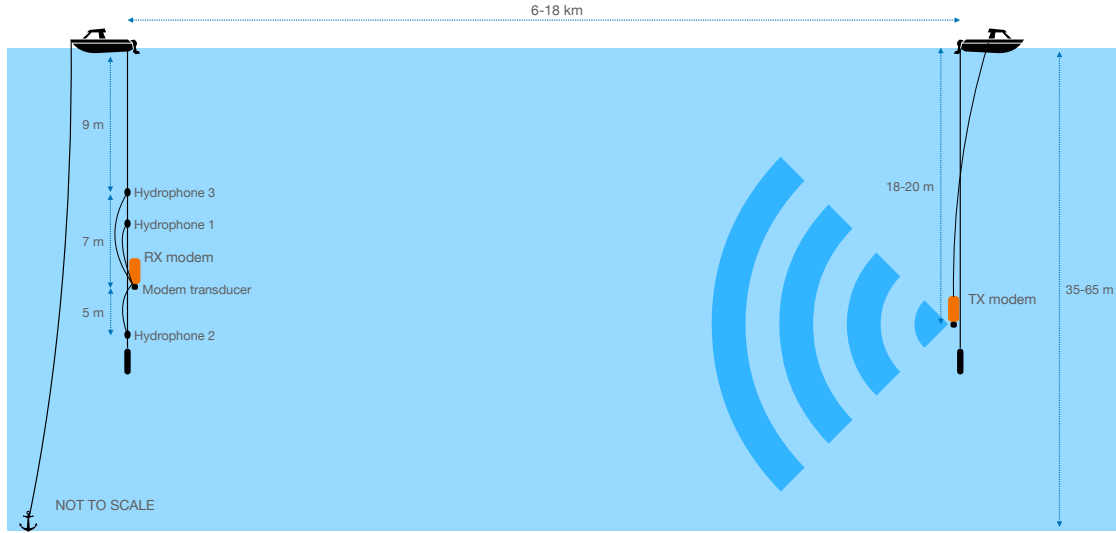


Figure 2.4: An illustration showing the transmitter and receiver setup for the experiment (not to scale).

and connected to a laptop. The modem was deployed at about 18–20 m depth during times of low current, but might have been significantly shallower during phase 2 when the current was strong. The overall setup is illustrated in Figure 2.4.

Four different types of communication signals in the 4.5–5.5 kHz band were transmitted over the course of the experiment:

1. **FH-BFSK1**: FH-BFSK with 5 hops and a (31, 6, 7) BCH code, as recommended in deliverable D3. Each frame contains 17 bytes of user data and offers an information rate of about 18 bps.
2. **FH-BFSK2**: FH-BFSK with 10 hops and a (31, 11, 5) BCH code, as recommended in deliverable D3. Each frame contains 21 bytes of user data and offers an information rate of about 17 bps.
3. **JANUS³**: FH-BFSK with 13 hops as recommended in the NATO JANUS specifications and out-of-the-box supported by Subnero modems, but shifted to operate in the 4,500–5,540 Hz band. The data is protected by a 1/3-rate LDPC code, and each frame contains 17 bytes of user data, offering about 12 bps of information rate.
4. **OFDM**: 128-carrier OFDM with BPSK modulation on each carrier and a cyclic prefix of 43 ms. The data is protected by a 1/6-rate LDPC code, and each frame contains 64 bytes of user data, offering about 100 bps of information rate. This scheme is out-of-the-box supported by Subnero modems.

Days 1–2 focused on FH-BFSK1 and FH-BFSK2 at several ranges. Day 3 was used to transmit all 4 signals at various ranges. It was determined that FH-BFSK2 had the most robust performance of all tested

³While we use the name JANUS here to refer to a FH-BFSK with the JANUS hopping sequence, this scheme does not comply with the JANUS specifications in terms of frequency band, detection preamble, error correction coding or frame format. We only use the name JANUS to differentiate it from the other FH-BFSK schemes designed in this project and detailed in deliverable D3.

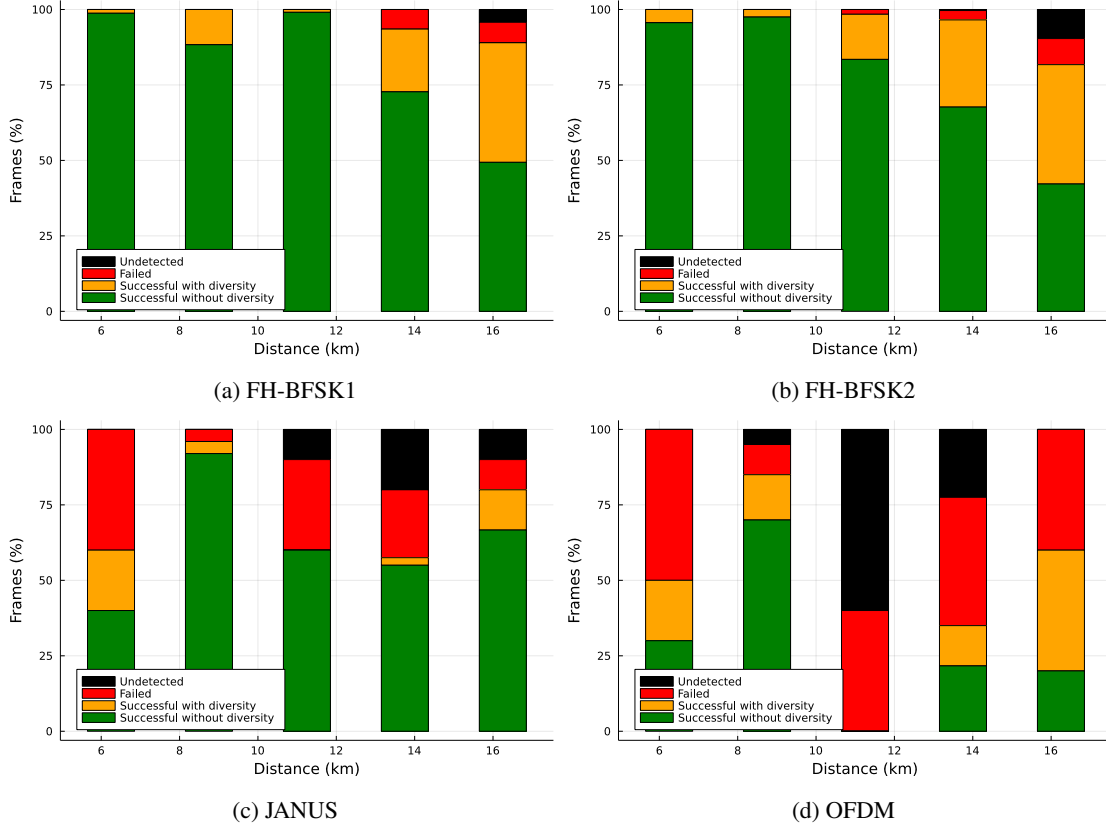


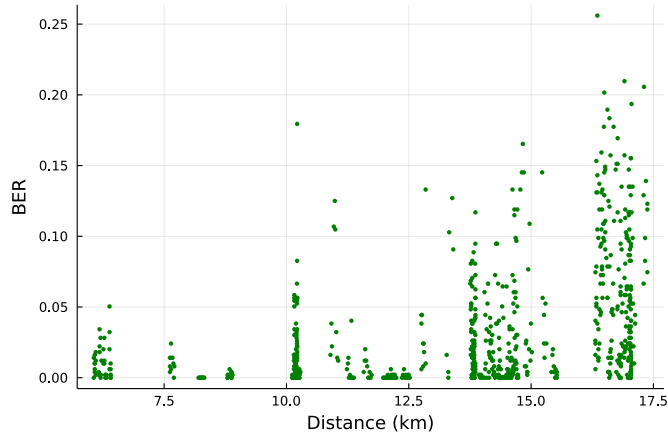
Figure 2.5: Communication performance summary for FH-BFSK1, FH-BFSK2, JANUS and OFDM frames over the 5 days and various transmission ranges from 5 to 17.5 km.

schemes during days 1-3, and the frames were larger and hence more suitable for larger data transfer. We therefore used FH-BFSK2 for the demonstration of larger data transfer tasks on day 4-5.

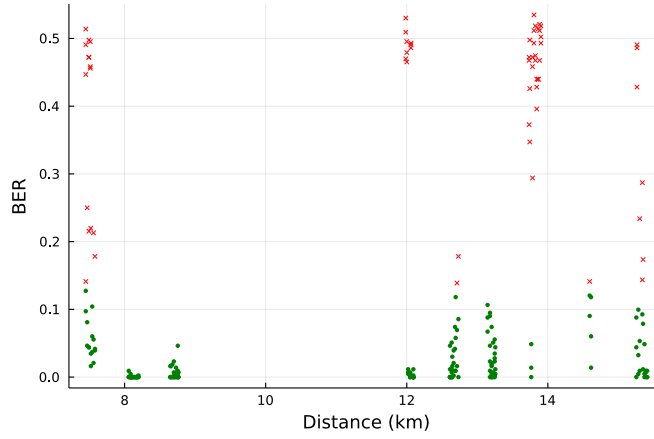
2.2 Communication performance summary

We begin this section by defining some terminology that we will need throughout the section, and in later parts of the report:

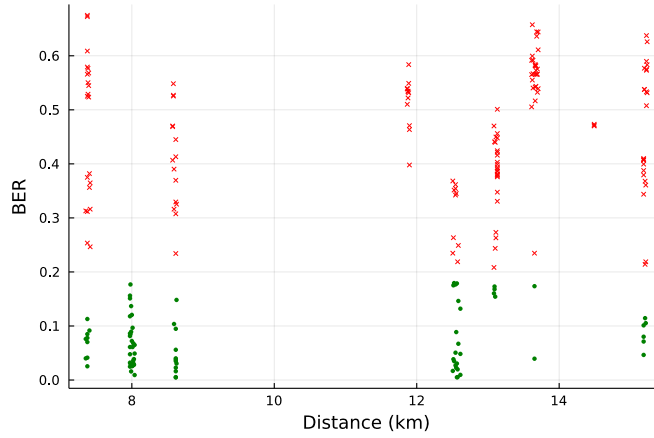
- **Undetected frame:** Frames are transmitted with a detection preamble. This detection of this preamble at the receiver triggers an acoustic capture and processing of the frame. When the channel conditions are very poor, a transmitted frame may go undetected at the receiver, and therefore results in frame loss.
- **Successful frame (without diversity):** Once a frame is detected, it is demodulated and decoded. A CRC check is performed to check the frames for errors. If no errors are present, the frame is considered to be successfully received without diversity.
- **Successful frame (with diversity):** If errors are detected during the CRC check, spatial diversity combining is attempted, and the combined frame is decoded and the CRC check is attempted again.



(a) FH-BFSK1 & FH-BFSK2



(b) JANUS



(c) OFDM

Figure 2.6: BER performance summary for FH-BFSK1/FH-BFSK2, JANUS and OFDM frames. Each green dot represents one successful reception of a frame at the given range, whereas a red cross represents a failed frame. Undetected frames are not seen in these plots. For FH-BFSK1/FH-BFSK2, we do not have BER estimates for the few failed frames, and so no red crosses are shown; however, most of the frames were successfully received and seen as green dots.

If no errors are present after combining, the frame is considered to be successfully received with diversity.

- **Failed frame:** If the frame still has errors after diversity combining, it is considered a failed frame and contributes to frame loss.
- **BER:** The bit error rate (BER) refers to the ratio of the number of bits in error to the total number of bits in the frame *before error correction decoding*. If decoding is successful, the number of errors after decoding reduces to zero.

With this terminology in place, we next look at a summary of communication performance (Figure 2.5) over the 5 days at various transmission ranges. The green bars denote the percentage of frames that were successfully decoded using a single hydrophone at the receiver. This provides an estimate of single transducer modem performance. The orange bars show the additional percentage of frames that were decoded successfully through spatial diversity processing, i.e., combining information from the 4 receive channels (1 transducer + 3 hydrophones). The red bars show the percentage of frames that were detected but could not be recovered without errors (“failed frames”), and hence had to be discarded. The black bars show the percentage of frames that were transmitted but could not be detected at the receiver.

Detection performance does not depend on the communication scheme, and so poor detection performance is indicative of poor propagation conditions in the channel at the time the data was collected. Keeping this in mind, the poor performance of OFDM at 11–12 km range should not be regarded as a shortcoming of OFDM, but rather an indicator of a poor channel during that data collection. A large percentage of failed frames, on the other hand, likely indicates lack of robustness of that communication scheme.

From Figure 2.5, we see that the performance of FH-BFSK1 and FH-BFSK2 at all ranges is very good (>80% frame delivery rate) with spatial diversity. Even without using spatial diversity processing, the performance is reasonably good (>70% frame delivery) up to about 14 km, and gracefully degrades beyond that. There is no significant difference in overall performance between FH-BFSK1 and FH-BFSK2⁴. On the other hand, the JANUS scheme somewhat worked at most ranges, but yielded less robust performance and offered a lower data rate. The OFDM scheme offered a 5× data rate advantage, but only worked well when the channel conditions were favorable and ranges short.

To understand the performance further, in Figure 2.6, we plot the bit error rate (BER) of the various communication schemes as a function of communication range. We see that for FH-BFSK1/FH-BFSK2, the BER steadily becomes more variable as communication distance increases. The schemes are well protected by the BCH code, making them robust to small changes in channel or noise conditions, and therefore we see that most of the frames have very few errors. The BCH codes we are using for FH-BFSK1 and FH-BFSK2 are able to correct errors up to a BER of approximately 0.2. When JANUS works well, it also has relatively small BER. However, when the channel conditions are not favorable, JANUS BER performance is much poorer. The OFDM BER is rarely close to zero, and hence the LDPC error correction code is working hard even in good channel conditions to recover the transmitted data. When the channel or noise conditions deteriorate, the BER increases and OFDM quickly fails. This is not surprising, as phase coherent communication is known to be very sensitive to channel conditions, and our previous channel studies (presented in deliverable D2) showed that phase stability at long ranges is generally poor. In summary, FH-BFSK1 and FH-BFSK2 were designed for robustness, and they indeed demonstrate robust performance at all ranges. JANUS and OFDM implementations in the Subnero modems, on the other hand, were designed for shorter range mid-frequency communications. While they work at longer

⁴On days 1–3, we found FH-BFSK2 to perform somewhat better than FH-BFSK1, but with further data collection on days 4–5, that difference no longer seemed significant. We therefore conclude that both schemes are similarly effective, when averaged across the data collected over the entire experiment.

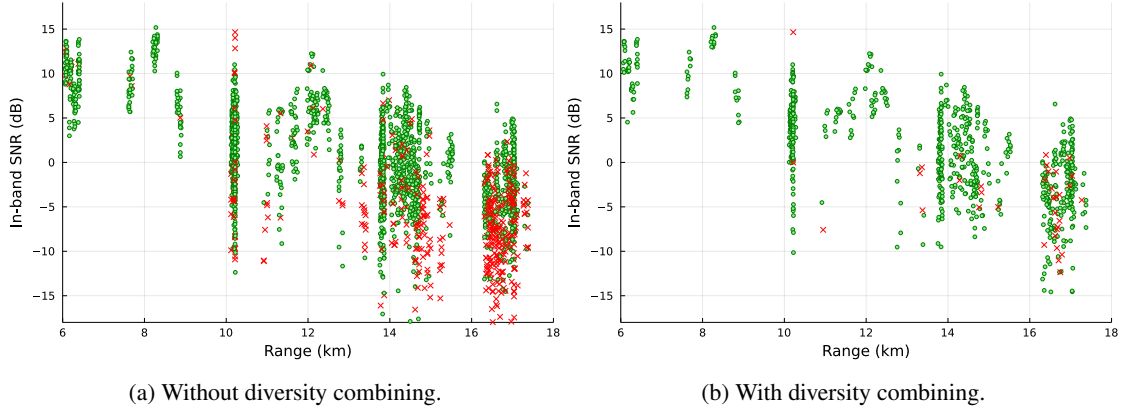


Figure 2.7: SNR as a function of transmission range. A green dot represents a successful frame reception, while a red cross indicates a failed frame.

ranges and lower frequencies when the conditions are favorable, they are not as robust as FH-BFSK1 and FH-BFSK2.

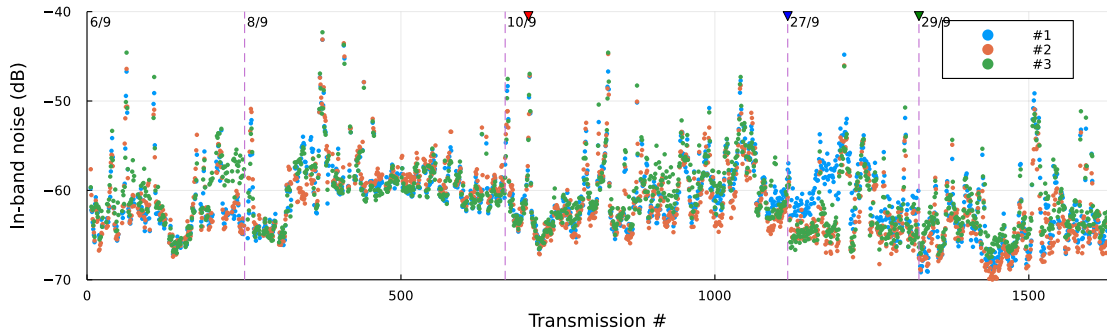
2.3 Channel and noise variability

From Figure 2.6, we see a very large variability in BER performance, especially at longer ranges. What causes it? Is it due to variability in ambient noise, or in transmission loss, leading to a variability in signal-to-noise ratio (SNR)? To answer these questions, we analyze ambient noise at the receiver as a function of range and time, over the period of the experiment. To avoid being contaminated by our transmissions, we pick a 4-second window just prior to the reception of each of our own transmissions to measure ambient noise. We can be sure that our transmissions do not arrive at the receiver during that period, as we honor a 50% duty cycle and leave a large gap before each transmission. We also measure received signal strength indicator (RSSI) from a 4-second window during our reception, and use it to estimate SNR.

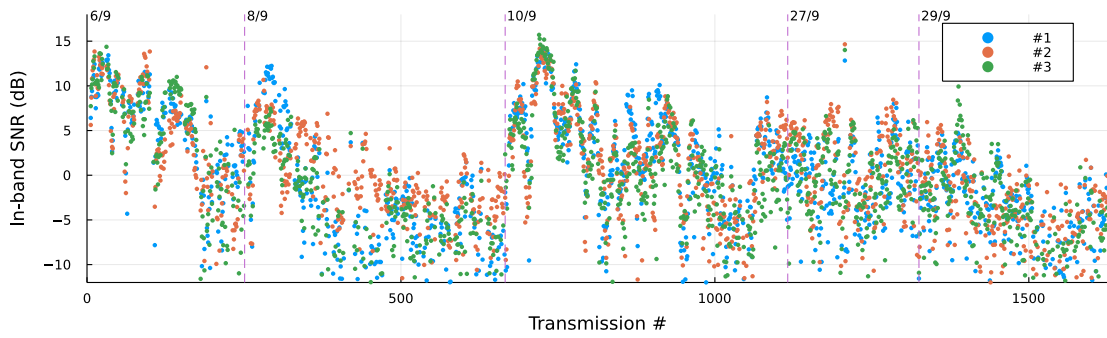
In Figure 2.7, we plot the SNR as a function of transmission range. The average SNR clearly reduces as a function of range, but we see a lot of variability in SNR and most of the failed frames (red crosses) are due to low SNR. While diversity combining improves the SNR and consequently reduces frame failure, the combined SNR still shows significant variability. In order to understand where this variability comes from, we look at the ambient noise and SNR as a function of time.

Figure 2.8 (a) shows how ambient noise varies over the duration of the experiment. Each point in the plot is an ambient noise measurement during a period just before the reception of a frame. The plot shows a 15–20 dB variability in noise level, with short spikes that can even exceed 20 dB. We pick three representative times of low (green arrow), medium (blue arrow) and high (red arrow) ambient noise, to see what might be causing the noise. The spectrograms of the noise at these three times are shown in Figure 2.9. From the spectrogram, and by listening to the recordings, it is clear that the noise is caused due to passing ships. Given the busy Singapore shipping lane next to our receiver station, this is not surprising.

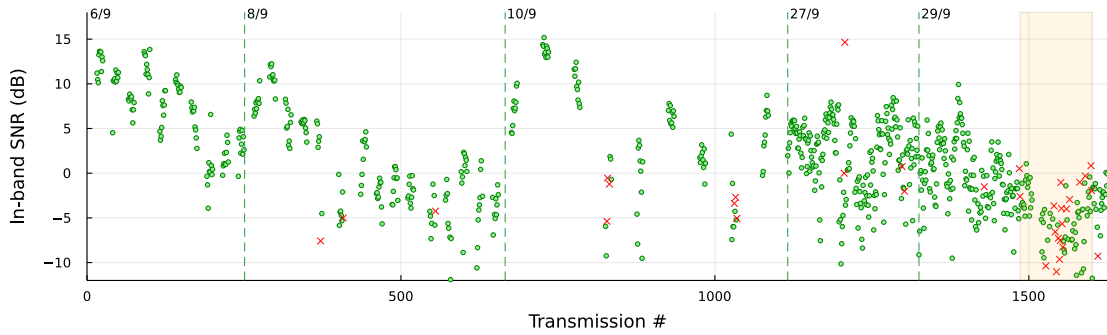
In order to understand whether the SNR variability is mostly due to ambient noise, in Figure 2.8 (b), we plot the SNR over the duration of the experiment. By comparing it to Figure 2.8 (a), we see that quite a lot of the SNR variability is indeed due to ambient noise changes, but not all. For example, right at the end around transmission #1600, we have a time of low SNR and low ambient noise. This also corresponds to a time of high frame loss, as we see in the highlighted period in Figure 2.8 (c). It seems that during this



(a) Ambient noise level, measured in a 4-second window just before each received frame.



(b) SNR, with noise measured in a 4-second window just before each received frame and RSSI measured in a 4-second window during the frame.



(c) SNR for the best hydrophone, with a green dot indicating a successful frame and a red cross indicating a failed frame.

Figure 2.8: Ambient noise level (with reference to full scale ADC output) and SNR over time, on hydrophones #1–#3. The vertical dashed lines denote breaks between experiment days (dd/mm labeled). The red, blue and green arrows at the top indicate times of high, medium and low ambient noise respectively, with spectrograms at each time shown in Figure 2.9.

time, the signal strength was low, although the ambient noise was not unusually high. This suggests that the transmission loss has somehow increased significantly during that period.

We plot ship tracks using AIS data during the period of anomalous transmission loss in Figure 2.10. The likely reason for the increase in transmission loss is clearly visible as a ship track along our line

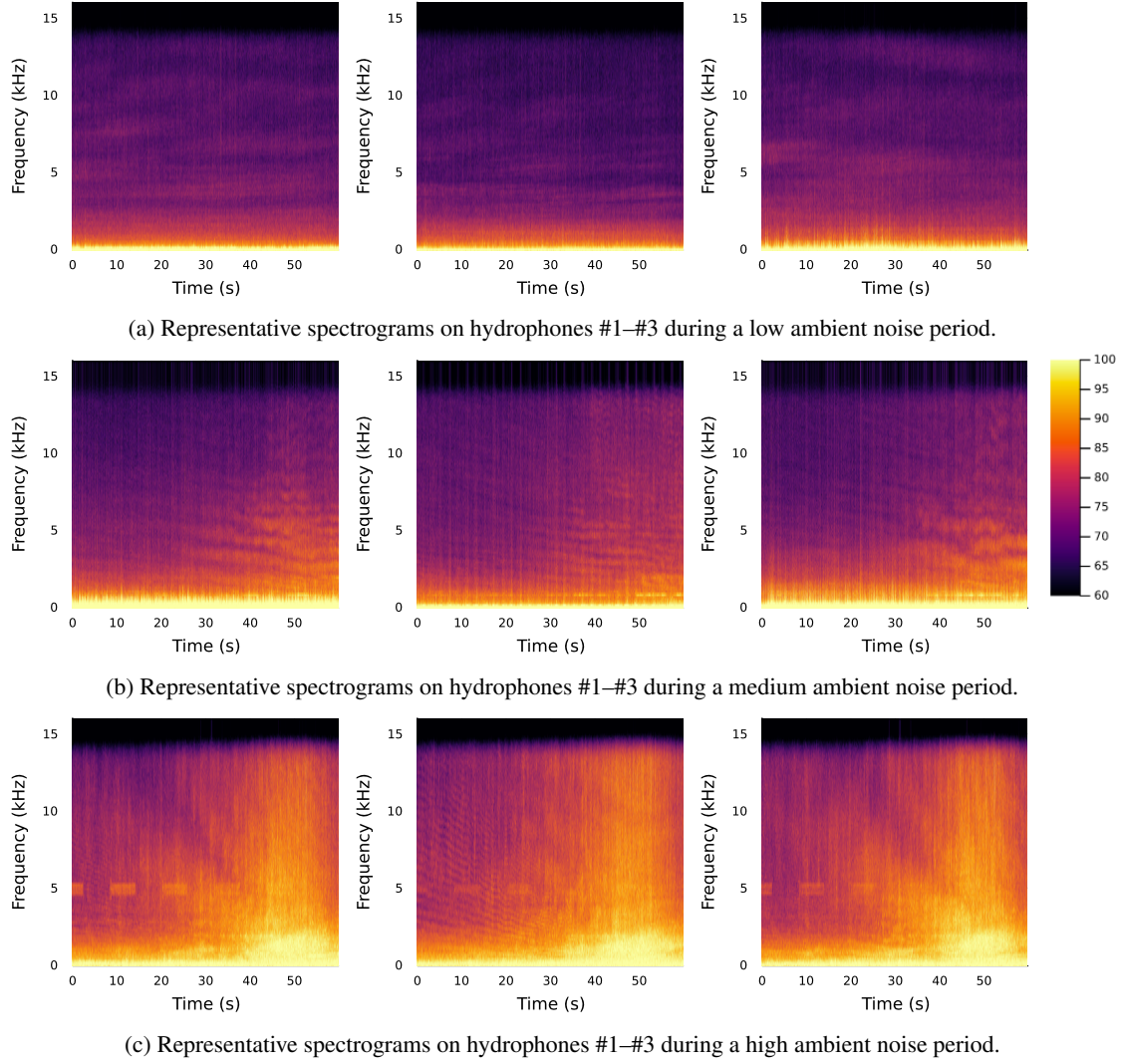


Figure 2.9: Spectrograms of ambient noise at 3 different times marked in Figure 2.8 (a) as a green arrow (low ambient noise), blue arrow (medium ambient noise), and a red arrow (high ambient noise). The colorbar is in dB re $1 \mu\text{Pa}/\sqrt{\text{Hz}}$, but should be considered approximate as the hydrophones are not calibrated.

of transmission, near the eastern end. This was a ship traveling directly along the line of sight from the transmitter to the receiver, and injecting large number of bubbles in the water due to its propeller. These bubbles are a serious impediment to acoustic transmission and therefore lead to a temporarily high transmission loss. The large bubbles are rapidly lost from the water column due to buoyancy, and so the transmission loss recovers to its original value and communication performance is restored shortly after the ship passes.

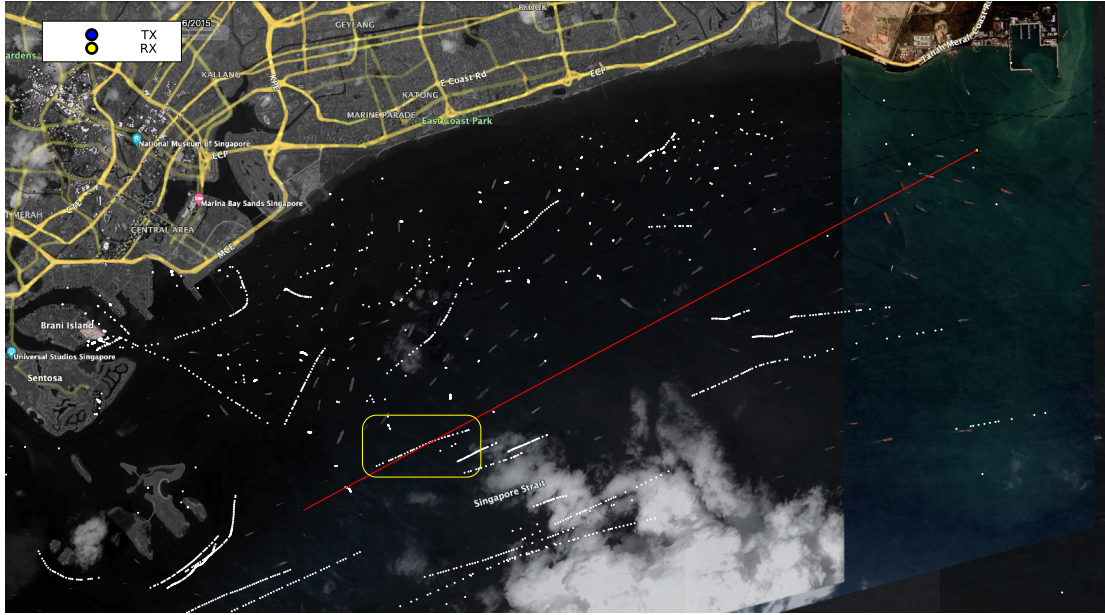


Figure 2.10: Ship tracks from AIS during a short period around transmission #1500–#1600. A ship track along the line of transmission is highlighted, and the likely cause for communication outage during this period.

2.4 Communication of larger datagrams

Most modern communication systems *fragment* data streams or large datagrams into smaller fixed-length frames for transmission, and *re-assemble* them at the receiver. The advantage of this approach is that when transient noise or poor channels conditions cause data to be lost, only a few frames are lost, and they can be re-transmitted if necessary. This approach is known as Automatic Repeat reQuest (ARQ). For ARQ to work, a feedback channel is needed to let the transmitter know *which* frames were lost. ARQ can therefore only be used when duplex communication is possible. Even with duplex communication, ARQ is very inefficient at long ranges in water, as the round-trip propagation delay at each cycle of the ARQ protocol can be in the order of tens of seconds.

To solve this problem, we had previously proposed the use of a class of codes known as *rate-less erasure correction codes* (sometimes also called *fountain codes*) in underwater communication systems [1]. While the approach proposed in that paper was directed towards improving the efficiency of ARQ, it can also be used to communicate reliably in a simplex scenario, as we demonstrated later in this section. Rate-less codes allow the creation of a virtually infinite sequence of frames that can be transmitted, deriving from a finite amount of data. As long as at least a certain number of frames are successfully received, the data can be recovered at the receiver. Based on a statistical estimate of frame loss probability, one can transmit sufficient frames to ensure that the minimum number of frames for successful recovery are received with a high probability. In deliverable D3, we presented a simple analysis to guide the choice of number of frames to transmit, and we will revisit this question again in Chapter 3 of this report. In the rest of this section, we will focus on how we used rate-less codes during the field demonstration to transmit large datagrams successfully over ranges up to 17.6 km underwater.

UnetStack uses specific rate-less erasure correction codes known as *Raptor codes* [2] in its datalink protocol. We leverage this implementation to provide us fragmentation and re-assembly of large data-

Transmission chain:



Reception chain:



Figure 2.11: Block diagram showing the transmitter and receiver processing chains for large datagram transfer.

grams, while at the same time providing robustness through erasure correction. The steps involved in data transfer of large datagrams are summarized in Figure 2.11, and described in detail next:

Transmitter:

1. The datagram to be transmitted is first compressed using the *gzip* DEFLATE algorithm (a combination of LZ77 and Huffman coding) [3]. If the compressed datagram is smaller than the original, we replace the datagram to be transmitted with its compressed version.
2. The datagram is then fed to a Raptor encoder, which creates a potentially infinite stream of fragments of a specified size from it. For the purposes of the field demonstration, we create 100 fragments of up to 18 bytes each from each of our test datagrams.
3. We package the Raptor-coded frames with some *side information* required for decoding. With the knowledge that the total number of fragments to transmit for any given datagram was not going to exceed 256, we are able to fit the side information in a 3-byte header. We therefore encode each Raptor-coded fragment in one FH-BFSK2 frame of 21 bytes (+ 1 byte CRC for error detection)⁵.
4. Each packaged frame is encoded using a (31, 11, 5) BCH code and then modulated using FH-BFSK2 modulation and transmitted.

Receiver:

1. At the receiver, incoming frames are detected, demodulated and decoded using the BCH decoder.
2. For each received frame, the CRC byte is used to check for errors. If any errors are found, the received frame is discarded. If no errors are detected, the frame is passed to the Raptor decoder.
3. The Raptor decoder checks the side information in the frame, and adds the fragment to its internal cache. If sufficient frames have been received to decode the datagram, the Raptor decoder generates a datagram and passes it to the decompression algorithm. Otherwise it waits for more frames to arrive.

⁵This represents a modest $3/18 = 17\%$ overhead to support fragmentation, re-assembly and robustness to frame loss.

Table 2.1: Summary of test datagrams transfers on day 4 and day 5.

Day # (date)	Transfer #	Range (km) (nominal)	Test Datagram ID	Size (bytes)	Compressed (bytes)	Min. number of fragments
4 (27/9)	1–9	10	rkif1	353	210	12
4 (27/9)	10–19	15	art3	509	185	11
4 (27/9)	20–21	15	art4	409	104	6
5 (29/9)	1–9	14	rkif1	353	210	12
5 (29/9)	10–21	17	art2	601	130	8
5 (29/9)	22–25	17	rkif4	363	252	14

4. On receiving a datagram from the Raptor decoder, the decompression algorithm checks its format to see if it has been compressed using gzip. If compression is detected, the gzip decompression algorithm is applied to decompress the datagram. Otherwise the incoming datagram is passed on unchanged to the output of the decompression step.
5. The resulting datagram is the received copy of the transmitted datagram. For the purposes of our demonstration, we timestamp and log the recovered datagram at the receiver.

Although the typical 450–600 byte test datagrams (130–252 bytes after compression) required only 10–15 frames to transmit, we transmitted about 100 frames each to collect statistical data on how many frames were required for successful reception of the datagram. The transmitter was instructed to discard its cache of received fragments each time a successful reception was completed and logged, and start a new reception. With this, we were able to transmit the same test datagram several times in a row and collect statistics for each successful delivery.

We summarize the test datagram transfers on day 4 and day 5 of the experiment in Table 2.1, as well as in Figures 2.12 and 2.13. Each horizontal *life-line* in the figures represents one datagram transfer with time running from left to right. A solid circle represents a successful delivery of a frame, whereas a cross represents a lost frame. When sufficient frames are received, the transfer is completed and marked as “DONE”, and the size in brackets is the datagram size after compression. Transfers marked as “ABORT” were stopped before sufficient frames were received, either because we reached the maximum number of transmissions (about 100) for that test, or we had to temporarily interrupt the transmissions due to logistical considerations (e.g. having to reposition RV Galaxea due to safety issues). We reproduce sample excerpts of the receiver logs for each of the test datagrams transferred in Appendix A.

We also annotate each successful transfer in Figure 2.12 and 2.13 with an effective information rate, defined as:

$$\text{Information rate} = \frac{\text{Number of user bits in datagram (after compression)}}{\text{Total transmission duration}}, \quad (2.1)$$

where the transmission duration is the sum of frame durations for all transmitted frames for the data transfer, i.e., the total time during which the transmitting modem is transmitting. In practice, we had to leave gaps between transmissions to honor the 50% duty cycle requirement for the transducer used in the Subnero modems. However, this requirement can easily be worked around by using a larger transducer, lower transmit power, or using two modems for transmission (frames transmitted alternately from each modem).

On day 4, communication at both (10 km and 15 km) ranges was stable and most frames were successfully delivered. Only 2 frames out of 108 frames transmitted at 10 km range were lost, and test datagrams were successfully delivered 8 times. The last delivery was aborted as we stopped transmission and moved

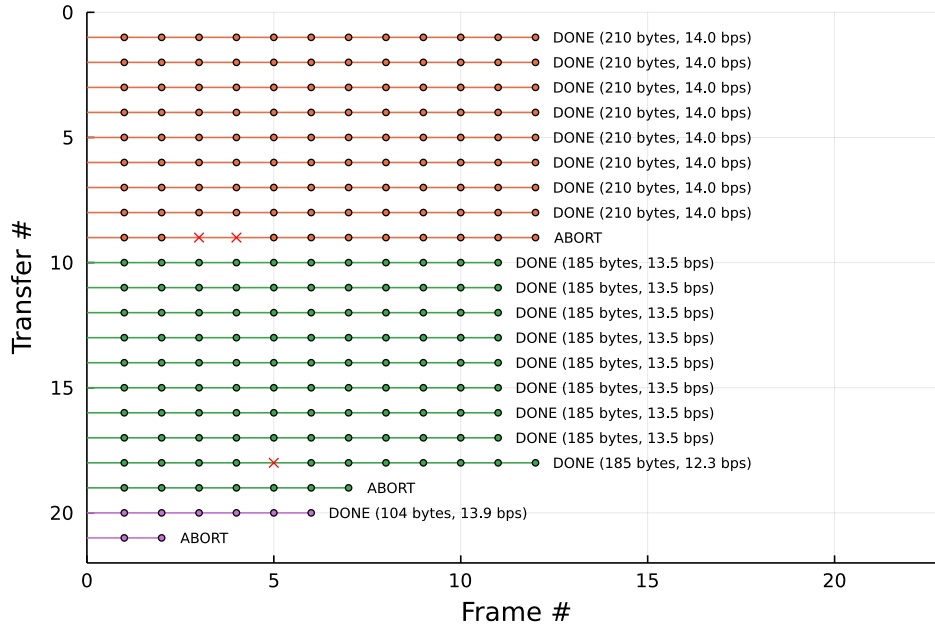


Figure 2.12: Summary of 21 test datagram transfers on day 4 (27/9). Solid circles represent successful frames, and crosses represent lost frames. Transfers #1 to #9 were 353 byte test datagrams (210 bytes after compression, minimum 12 fragments required for recovery) sent from a nominal range of 10 km, #10 to #19 were 509 byte test datagrams (185 bytes after compression, minimum 11 fragments required for recovery) sent from a nominal range of 15 km, and #20 to #21 were 409 byte test datagrams (104 bytes after compression, minimum 6 fragments required for recovery) sent from a nominal range of 15 km. The effective information rate (i.e., the data rate achievable if a modem transmitted successfully with a 100% duty cycle) for each data transfer is also annotated in case of successful transfers.

to 15 km range. Only 1 frame out of 115 frames transmitted at 15 km range was lost, and test datagrams were successfully delivered 10 times. Two of the deliveries were aborted, once to change the test datagram being transmitted, and the second because the receiver ran out of battery.

On day 5, communication at 14 km was stable, with 2 frames lost out of 107 frames transmitted. The communication at 17 km was initially stable, with only 3 frames lost out of 91 frames transmitted until end of transfer #20. As outlined in section 2.3 and Figure 2.10, bubbles from a ship traveling along the line of sight between the transmitter and receiver disrupted communication temporarily after that. We see this as a large number of lost frames during transfers #21 and #22. Transfer #21 was aborted as we stopped the transmission to change the test datagram, but transfer #22 was eventually successful when sufficient number of frames were received. The communication performance slowly recovered after that, with several lost frames during transfers #23 and #24, but successful delivery of datagram for both. Transfer #25 saw a full recovery of the channel, but was aborted as we ended the experiment and returned to land.

With the exception of day 5 transfers #21–#24, when we had a known reason for poor frame delivery performance, we can see from Figures 2.12 and 2.13 that it was sufficient to transmit 2 extra fragments per datagram transfer to guarantee successful delivery. In most transfers, the minimum number of fragments was sufficient, but the occasional frame loss could be protected against by transmitting 2 extra fragments for datagrams that were few hundred bytes in size. Transmitting even more extra frames when circum-

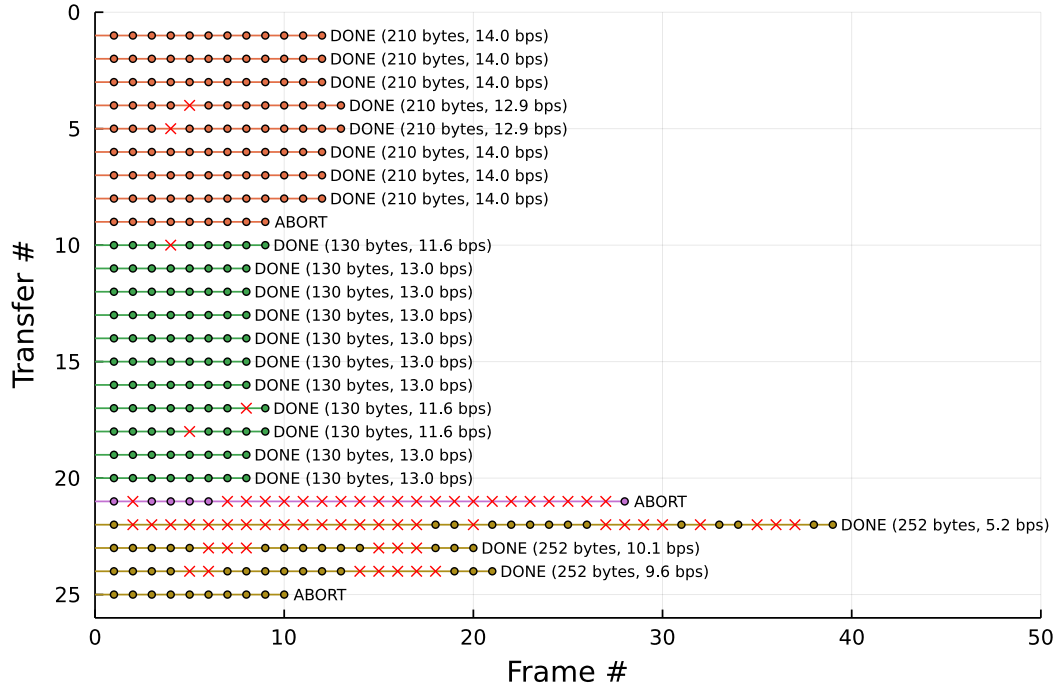


Figure 2.13: Summary of 25 test datagram transfers on day 5 (29/9). Solid circles represent successful frames, and crosses represent lost frames. Transfers #1 to #9 were 353 byte test datagrams (210 bytes after compression, minimum 12 fragments required for recovery) sent from a nominal range of 14 km, #10 to #21 were 601 byte test datagrams (130 bytes after compression, minimum 8 fragments required for recovery) sent from a nominal range of 17 km, and #22 to #25 were 363 byte test datagrams (252 bytes after compression, minimum 14 fragments required for recovery) sent from a nominal range of 17 km. The effective information rate (i.e., the data rate achievable if a modem transmitted successfully with a 100% duty cycle) for each data transfer is also annotated in case of successful transfers.

stances permit or reliability needs warrant, provides further protection (e.g., day 5 transfers #22–#24 were successful in spite of a large number of lost frames as we transmitted sufficient extra frames).

2.5 Summary

We tested the communication schemes proposed in deliverable D3, as well as those available off-the-shelf in Subnero modems at ranges from about 5 to 17.6 km. While all schemes worked well, not surprisingly, we found that the two schemes we designed for long-range communications based on the channel measurements in deliverable D2 performed more robustly. The key features of the two schemes that contribute to their good performance are summarized below:

- Use of incoherent frequency spread modulation (FH-BFSK) to provide robust communication (same as JANUS).
- Use of a periodic hopping sequence (rather than a pseudo-random hopping sequence that JANUS uses) to increase the time to re-use of frequency, thereby improving delay spread tolerance.

- Careful selection of number of hops, frequency spacing, and symbol duration based on constraints imposed by channel measurements (delay spread and Doppler spread).
- Use of interleaved BCH codes with a ordered statistics decoding (OSD) algorithm for effective error correction of short frames. Codes such as LDPC (used in Subnero modems) are good for large frames, but not as effective on short frames.

While there was no significant difference between the performance of both schemes, scheme FH-BFSK2 offers a larger frame size and hence is better suited for large datagram delivery. We therefore used FH-BFSK2 to demonstrate delivery of larger datagrams (few hundred bytes in size) through realtime fragmentation, re-assembly and erasure correction coding for robustness to occasional frame loss. 40 large datagrams were successfully delivered at ranges between 10–17 km on days 4 and 5 of the experiment as a demonstration.

CHAPTER 3

PERFORMANCE PREDICTION TOOL

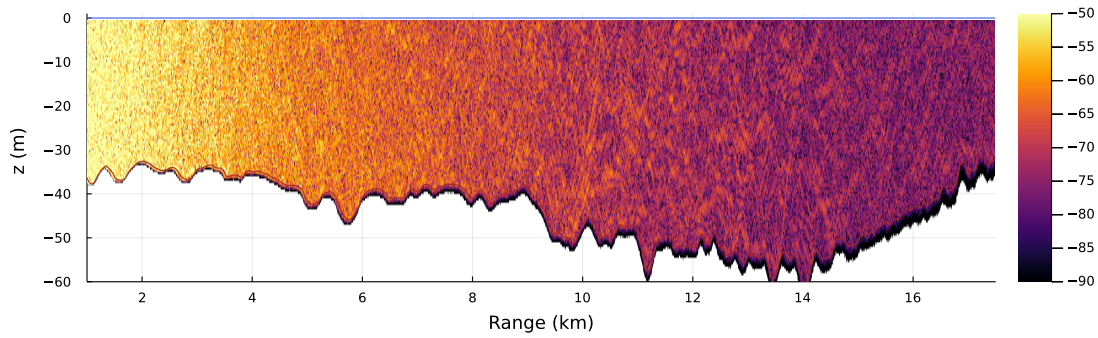
An acoustic propagation and communication performance modeling tool (`lrsim`) was developed in the early stages of this project to support simulation studies. With the communication scheme (FH-BFSK2) now finalized and tested, the tool has been upgraded to predict the performance of FH-BFSK2 in realistic channel conditions. In addition, the tool now also handles spatial diversity combining, as we found this to be valuable for the communication system. In this chapter, we compare the predictions of the upgraded tool against experimental findings from the last chapter.

3.1 Acoustic modeling

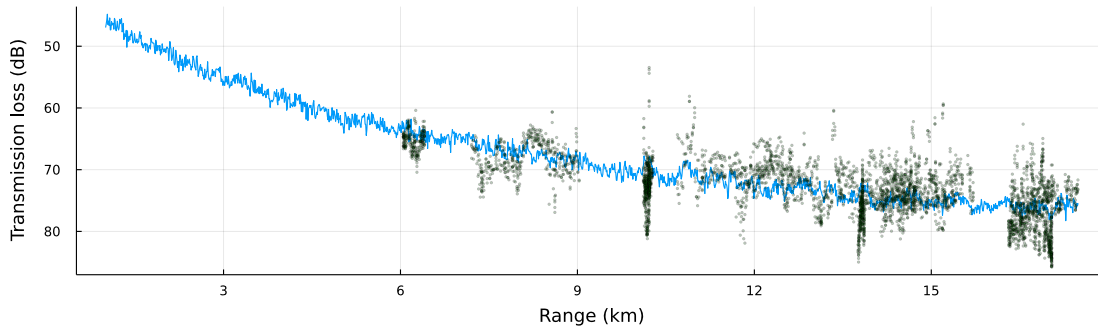
The `lrsim` tool utilizes the BELLHOP acoustic propagation model as a basic building block. BELLHOP is repeatedly used with a Monte Carlo sampling approach to obtain statistical predictions of communication performance. The Monte Carlo sampling method works by drawing samples from each of the distributions of various input parameters (e.g. transmitter depth, receiver depth, transmission range, etc), running BELLHOP acoustic propagation models for each set of samples, and then estimating the output distribution based on the resulting predicted transmission loss samples.

We start off by looking at a representative output from one nominal BELLHOP run in Figure 3.1. The key parameters used in the environmental description for the acoustic propagation model are listed below:

- Isovelocity sound speed profile with a nominal sounds speed of 1541 m/s, in alignment with CTD measurements shown in Figure 2.3.
- Bathymetry based on MPA charts at nominal tidal conditions, as shown in Figure 2.2.



(a) Predicted transmission coefficient (negative transmission loss in dB) shown as a function of transmitter depth and range from the receiver. The receiver is placed at a depth of 20 m.



(b) Averaged transmission loss (blue curve) over 10–30 m of transmitter depth, as a function of range from the receiver. The green markers are transmission loss estimates obtained from measurements during the experiment.

Figure 3.1: Predicted transmission loss from a BELLHOP simulation run for a nominal operating conditions in the area of interest, compared against estimates from experimental measurements.

- Seabed parameters consistent with *coarse sand* [4]: relative density of 2.231, relative sound speed of 1.2503, and attenuation 0.01638. While these values are consistent with the type of seabed known to be found around the area of operation, the choice was also guided by the desire to match the model output with experimental observations.

In Figure 3.1 (a), we see transmission loss of -70 to -80 dB at a range of about 15 km. There is significant variability in transmission loss estimates, with the estimate varying as much as 10–15 dB with only minor changes in transmitter and receiver location, or other environmental factors (e.g. bathymetry, sound speed, etc). This variability is qualitatively consistent with the variability in RSSI and SNR that we observed in our experiments (e.g. Figures 2.7 and 3.1 (b)).

The blue line in Figure 3.1 (b) shows average predicted transmission loss over the 10–30 m depth as a function of range. We see that the predictions (blue line) are consistent with experimental observations (green dots). The experimental observations in this plot span several hours on 5 different days. To understand the variability in transmission loss better, we plot estimated transmission loss values from experimental measurements as a function of transmission number (time) in Figure 3.2. For every 10 minute interval, we run a set of Monte Carlo acoustic propagation models for the nominal conditions during that interval, and plot the nominal transmission loss (blue markers and line). The modeling tool also provides us with statistical variability estimates, and so we plot a 90% confidence interval around each predicted transmission loss. We see that the observed values are indeed within the confidence interval most of the

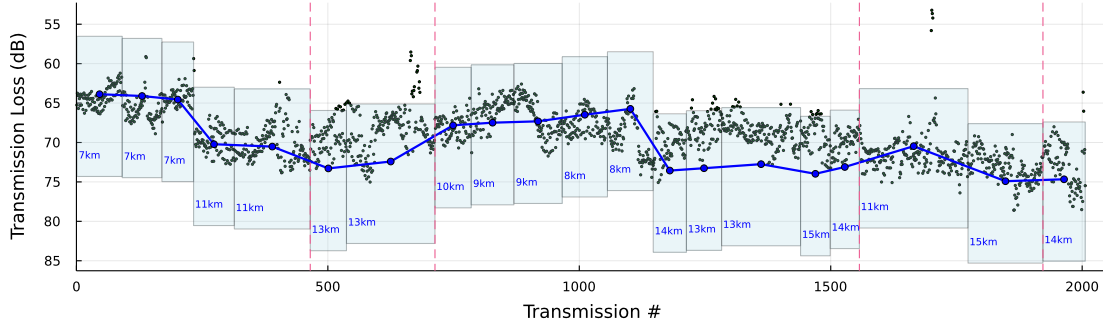


Figure 3.2: Comparison of modeled and measured transmission loss as a function of transmission number (time) spanning several hours each day, over 5 days of experiments at sea. The estimated transmission loss based on experimental measurements are shown as dark green markers. The predicted transmission loss values are shown as blue markers and line, with light blue rectangles denoting 90% confidence intervals.

time, and quite close to the nominal estimate often. We conclude that the acoustic propagation model is able to give us a good estimate of mean transmission loss as well as the variability in a shallow water environment such as the one we operated in Singapore waters.

3.2 Communication performance modeling

The acoustic propagation modeling described in the previous section not only yields statistical estimates of the transmission loss, but also of the channel impulse response. We use these impulse response estimates to simulate propagation of communication signals through the acoustic channel, and then add red¹ Gaussian noise to the received signals. By processing the signals using the algorithms used during the experiment on the modem, we get communication performance estimates. The `lrsim` tool also allows placement of multiple hydrophones at the receiver (receiver array). The received signals at each of the hydrophones is combined via selection diversity and log-likelihood ratio (LLR) combining, just as we did in the modems during the final experiment.

To check if the prediction tool was able to estimate communication performance well, we set up a simulation with the same parameters as the acoustic model in the previous section, along with the following additional parameters:

- Nominal ambient noise level of 80 dB re 1 $\mu\text{Pa}^2/\text{Hz}$. This noise level was chosen to cover noise levels of up to *medium* noise as shown in Figures 2.8 and 2.9. The power spectral densities measured on three hydrophones at low and medium noise conditions shown in Figure 3.3 suggest 80 dB to be a good upper estimate of noise.
- Transmitter depth of 19 ± 5 m (Gaussian distribution with mean of 19 m and standard deviation 5 m).
- Receiver depth (top hydrophone) of 9 ± 5 m (Gaussian distributed).
- Receiver array with 3 hydrophones. Second hydrophone 4 m below the top hydrophone, and the third hydrophone 8 m further below. This arrangement is similar to the one used during the experiment.

¹Red noise is defined as any noise whose power spectral density reduces by 6 dB per octave. This is consistent with observed power spectral density in most ocean conditions, and is therefore a good model for broadband ambient noise.

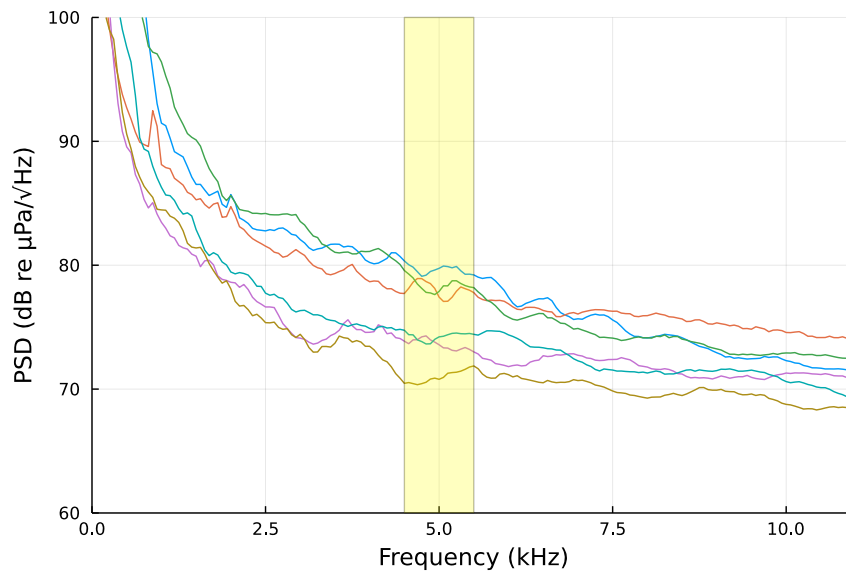


Figure 3.3: Power spectral density of ambient noise measured during the final experiment. The top 3 curves are measurements at the 3 hydrophones at medium noise conditions, and the bottom 3 curves are measurements at the 3 hydrophones at low noise conditions as denoted in Figure 2.8.

- Transmitter to receiver range Gaussian distributed around the nominal range of interest, with a standard deviation of 500 m.

The above parameters translate into an example simulation description (for a 15 km range) as shown below:

example-2.txt (15 km)

```
env = UnderwaterEnvironment(
    seasurface = Vacuum,
    ssp        = IsoSSP(1541.0),
    altimetry  = FlatSurface(),
    bathymetry = SampledDepth("bathy1.txt"),
    seabed     = CoarseSand,
    salinity   = 35.0,
    noise      = RGN(80.0)
)
txdepth = (19.0, 5.0)
rxdepth = (9.0, 5.0)
rxrange = (15000.0, 500.0)
rxarray = [
    (0.0, 0.0),
    (0.0, 4.0),
    (0.0, 12.0)
]
scheme = fhbfsk2
srclevel = 185.0
```

The output from the lrsim tool for the above simulation is shown next:

Simulation summary

Nominal transmission range: 15000.0 +/- 500.0 m
 Nominal transmitter depth: 19.0 +/- 5.0 m
 Nominal receiver depth: 9.0 +/- 5.0 m
 Number of hydrophones at receiver: 3
 Maximum water depth: 61.3 m
 Nominal frequency: 5000.0 Hz
 Nominal bandwidth: 1000.0 Hz

Propagation modeling

Nominal transmission loss: 71.7 dB
 50% confidence interval: [68.7, 77.0] dB
 90% confidence interval: [67.2, 79.7] dB

Nominal 6 dB delay spread: 106.6 ms
 50% confidence interval: [117.9, 92.7] ms
 90% confidence interval: [155.4, 84.9] ms

Communication performance

Communication scheme: FHBFSK2 (see LRComms D3)
 Frame length: 168 user bits (+ 8 bit CRC)
 Frame duration: 10.0 seconds
 User data rate: 16.8 bps
 Frame success rate: 100.0 %
 Diversity usage: 10.0 %
 Uncoded BER (min/median/max): 0.0 / 0.0 / 0.1653

Data transfer recommendations

Datagram size: 22 bytes
 Minimum number of frames: 13

# of frames transmitted	Probability of successful transfer	
	No diversity	With diversity
13	25.4%	100.0%
14	58.5%	100.0%
15	81.6%	100.0%
16	93.2%	100.0%
17	97.8%	100.0%
18	99.4%	100.0%
19	99.8%	100.0%
20	100.0%	100.0%

Table 3.1: Summary of performance prediction from the `lrsim` tool at various ranges, compared with experimental performance estimates. The overall frame success rate is specified, with the percentage of frames that required diversity processing shown in parenthesis.

Range (km)	Measured transmission loss (dB)	Predicted transmission loss (dB)	Measured frame success rate (diversity)	Predicted frame success rate (diversity)
10.0	70.0	68.3	99% (7%)	100% (0%)
15.0	73.7	71.7	96% (52%)	100% (10%)
17.5	75.6	74.3	96% (39%)	100% (27%)

We see that the `lrsim` tool predicts a transmission loss of around 71.7 dB, which is consistent with experimental observations (Figure 3.1 (b)). We see a 100% frame success rate, with 10% frames recovered with diversity processing (using data from multiple hydrophones). Comparing this with Figure 2.5 (b), we see reasonable agreement, with $> 95\%$ frame success rate in experiment at 14 km, and about 25% frames recovered with diversity processing.

We tabulate performance prediction results for 10 km, 15 km and 17.5 km in Table 3.1. Do note that the experiment saw plenty of shipping traffic affecting ambient noise and acoustic propagation, which we have not modeled in the performance prediction tool. Yet, the tool provides a useful estimate of transmission loss and communication performance at various ranges.

3.3 Other sample scenarios

DSO had requested communication performance for two sample scenarios to be evaluated:

1. 15 km range, with a single hydrophone receiver.
2. 17 miles (27 km) range, with a single hydrophone receiver.
3. 25 miles (45 km) range, with a 24 hydrophone vertical receiver array.
4. 25 miles (45 km) range, with a 24 hydrophone horizontal receiver array.

Two types of sound speed profiles and seabed types are used for the simulations:

- (ISO) Isovelocity sound speed profile with a nominal sound speed of 1540 m/s, or
- (SSP1) Constant sound speed of 1543 m/s up to a depth of 30 m, linearly reducing to 1536 m/s from 30 m to 100 m depth.
- (MUD) Seabed parameters consistent with a muddy bottom (*silty clay*): relative density of 1.146, relative sound speed of 0.9824, and attenuation 0.00163, or
- (SAND) Seabed parameters consistent with a sandy bottom (*coarse sand*): relative density of 2.231, relative sound speed of 1.2503, and attenuation 0.01638.

Other parameters used in all simulations are as follows:

- Constant bathymetry with a water depth of 100 m.
- Nominal ambient noise level of 55 dB re $1 \mu\text{Pa}^2/\text{Hz}$ (as advised previously by DSO).

Table 3.2: Summary of performance prediction results for FH-BFSK2 various scenarios of interest to DSO. The recommended number of frames shown is for 256 byte datagram delivery with 95% or better probability of success without diversity. When diversity is available, the second number (after the “/”) is the number of frames using diversity.

Range (km)	Speed of sound profile	Seabed type	# of Hydrophones	Transmission Loss (dB)	Frame success rate	Diversity usage	Recommended # of frames
15	ISO	MUD	1	80.5	100%	n/a	13
15	ISO	SAND	1	70.3	100%	n/a	13
15	SSP1	MUD	1	92.6	100%	n/a	13
15	SSP1	SAND	1	70.2	100%	n/a	13
27	ISO	MUD	1	89.4	90%	n/a	17
27	ISO	SAND	1	81.9	100%	n/a	13
27	SSP1	MUD	1	109.4	30%	n/a	inf*
27	SSP1	SAND	1	82.2	100%	n/a	13
45	ISO	MUD	24 vertical	102.6	99%	19%	20 / 14
45	ISO	SAND	24 vertical	94.0	100%	0%	13 / 13
45	SSP1	MUD	24 vertical	112.0	70%	70%	inf* / 24
45	SSP1	SAND	24 vertical	93.9	100%	0%	13 / 13
45	ISO	MUD	24 horizontal	100.5	90%	16%	22 / 17
45	ISO	SAND	24 horizontal	86.8	100%	0%	13 / 13
45	SSP1	MUD	24 horizontal	118.9	20%	20%	inf* / inf*
45	SSP1	SAND	24 horizontal	90.2	100%	10%	17 / 13

* infeasible with a reasonable number of frames.

- Transmitter depth of 5 ± 2 m.
- Receiver depth of 10 ± 5 m.
- For the case of 45 km range with a vertical array, we have a receiver array with 24 hydrophones spanning 10 m to 56 m water depth (± 5 m). For the 45 km range horizontal array, we have a receiver array with 24 hydrophones with a spacing of 10 cm between hydrophones. For all other cases, we have a single receiver hydrophone.
- Transmitter to receiver range Gaussian distributed around the nominal range of interest, with a standard deviation of 500 m.

The detailed results are included in Appendix B. We tabulate a summary in Table 3.2. We see good communication performance at all ranges when the seabed is sandy or we have an iso-velocity soundspeed profile. However, if we have a muddy bottom with a downward refracting soundspeed profile, most of the acoustic energy is lost in the seabed at very long ranges, and communication becomes challenging. Spatial diversity from the 24-element array helps improve the performance at 45 km range, allowing fair communication even in the most challenging conditions.

3.4 Summary

The updated `lrsim` communication performance prediction tool takes in an environmental description and makes prediction of transmission loss, impulse response and communication performance. Even with incomplete knowledge of the environment and shipping, the predictions from the tool can be quite useful. The measured transmission loss during the field experiment matched within about 2–3 dB of the predicted transmission loss, with appropriate choice of environmental parameters. The communication performance, measured in terms of frame delivery rate, also matched well between the tool's prediction and observations from field experiments.

CHAPTER 4

SUMMARY, CONCLUSIONS & RECOMMENDATIONS

In this project, we set out to develop technologies for robust medium and long -range communication. While the technology for mid-frequency medium-range communication has been developed and tested over several projects in the past years, the technology for low-frequency long-range communication was to be developed from ground up in this project. Thus the development of the low-frequency communication system became a focus area. We started off with a long-range acoustic propagation measurement exercise that gave us a good dataset for calibration of low-frequency propagation models, as well as for noise and channel variability modeling. We augmented this dataset with additional measurements made at St. Johns Island Netiquette setup, and developed a good understanding of low-frequency long-range acoustic propagation (transmission loss, impulse response, etc), variability (over short and long -term) and ambient noise. This informed the development of a set of communication schemes that were likely to provide robust performance at tens of kms of communication range. The key findings from this phase of the project are covered in deliverable D2, with some additional information in deliverable D3.

We worked with Subnero to implement the proposed schemes on low-frequency modems developed by them. In a series of field experiments conducted along the east coast of Singapore, we tested and demonstrated the proposed schemes at ranges of up to 17.6 km. Initial tests allowed us to pick the most promising scheme (dubbed FH-BFSK2), and test it further. We adopted techniques such as erasure correction codes for fragmentation and re-assembly, as they provided robustness to occasional frame loss and reliable message delivery even with a simplex communication link. This was demonstrated successfully at sea, at various ranges from 8 km to 17 km. The communication performance measurements from the experiment were compared against the predictions from the models developed in the early part of the project, and were found to match well. The models were encapsulated in a performance prediction tool

(called `lrsim`), packaged for use by DSO as desired. The results were presented earlier in this deliverable report (D4).

In parallel with the development of the low-frequency long-range communication system, we also pursued technologies to improve robustness of communication systems at all frequencies. One such technology is the use of spatial diversity to improve communication performance. We demonstrated that spatial diversity benefits both low-frequency and mid-frequency systems. In the long-range communication experiments, we used vertical spatial diversity effectively, almost doubling the number of frames successfully delivered during challenging channel conditions, as compared to using a single hydrophone for reception. With mid-frequency modems, we showed in deliverable D3 that a compact hydrophone array of just few tens of cms is sufficient to yield significant performance gains. In case of the low frequency modems, a spacing of few meters is desirable. We highly recommend that all future communication systems harness the power of spatial diversity when the application scenario permits it.

Another technology used to improve communication performance is link adaptation, provided a feedback channel is available in the form of a half-duplex link. By smartly adjusting parameters of a communication link, we are able to improve link data rate and robustness. Using the mid-frequency modems installed at the St. Johns Island Netiquette setup, we demonstrated a Gaussian process regression based algorithm that balances exploration of the link parameter space with exploitation of knowledge already gained, to obtain good link performance. This work is covered in detail in deliverable D3.

It is inevitable that communication outages sometimes occur, despite the best efforts to make communication systems robust. For example, we showed in Chapter 2 of this report that a ship traveling along the communication path and injecting bubbles in the communication channel, can temporarily disrupt communication. Disruption tolerant networks (DTNs) provide a way to communicate robustly in spite of these outages, as long as the data to be transferred is not time sensitive and a feedback channel is available (at least half-duplex links). In deliverable D3, we used a long-term network trace collected with the St. Johns Island Netiquette setup to show how DTN is able to spool data during an outage, and deliver it later when the link recovers. While the idea of a DTN is simple, its implementation requires careful coordination with different parts of the network stack. As part of this project, we developed a prototype DTN implementation on UnetStack – the network stack that powers all ARL and Subnero modems. This implementation is being incubated, and in the future may be made available for use by users of mid-frequency as well as low-frequency UnetStack-based modems.

APPENDIX A

SAMPLE LOGS OF TEST DATAGRAM TRANSFERS

day4/rx/log-0.txt (rkif1)

```
1632715465253|Reassembly complete:
```

Mon Sep 27 12:04:25 SGT 2021 (353 bytes)

[illegible]

If you can keep your head when all about you

Are losing theirs and blaming it on you,

If you can trust yourself when all men doubt you,

But make allowance for their doubting too;

If you can wait and not be tired by waiting,

Or being lied about, dont deal in lies,

Or being hated, dont give way to hating,

And yet dont look too good, nor talk too wise

[illegible]

day4/rx/log-0.txt (art3)

```
1632718559350|Reassembly complete:
```

Mon Sep 27 12:55:59 SGT 2021 (509 bytes)

[illegible]

day4/rx/log-0.txt (art4)

```
1632720540723|Reassembly complete:
```

Mon Sep 27 13:29:00 SGT 2021 (409 bytes)

[illegible]

day5/rx/log-1.txt (rkif1)

```
1632888070365|INFO|rx$1/yoda@49:doInvoke|Reassembly complete:
```

Wed Sep 29 04:01:10 UTC 2021 (353 bytes)

```
<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<
If you can keep your head when all about you
Are losing theirs and blaming it on you,
If you can trust yourself when all men doubt you,
But make allowance for their doubting too;
If you can wait and not be tired by waiting,
Or being lied about, don't deal in lies,
Or being hated, don't give way to hating,
And yet don't look too good, nor talk too wise
```

[illegible][illegible]

APPENDIX B

LRSIM OUTPUTS FOR VARIOUS SCENARIOS

15 km, ISO, MUD

Simulation summary

Nominal transmission range: 15000.0 500.0 m
Nominal transmitter depth: 5.0 2.0 m
Nominal receiver depth: 10.0 5.0 m
Number of hydrophones at receiver: 1
Maximum water depth: 100.0 m
Nominal frequency: 5000.0 Hz
Nominal bandwidth: 1000.0 Hz

Propagation modeling

Nominal transmission loss: 80.5 dB
50% confidence interval: [78.5, 84.4] dB
90% confidence interval: [77.1, 87.9] dB

Nominal 6 dB delay spread: 3.6 ms

50% confidence interval: [3.7, 3.5] ms
 90% confidence interval: [3.9, 3.3] ms

Communication performance

Communication scheme: FHBFSK2 (see LRComms D3)
 Frame length: 168 user bits (+ 8 bit CRC)
 Frame duration: 10.0 seconds
 User data rate: 16.8 bps
 Frame success rate: 100.0 %
 Diversity usage: 0.0 %
 Uncoded BER (min/median/max): 0.0 / 0.0 / 0.0

Data transfer recommendations

Datagram size: 256 bytes
 Minimum number of frames: 13

# of frames	Probability of successful transfer	
transmitted	No diversity	With diversity
13	100.0%	100.0%

15 km, ISO, SAND

Simulation summary

Nominal transmission range: 15000.0 500.0 m
 Nominal transmitter depth: 5.0 2.0 m
 Nominal receiver depth: 10.0 5.0 m
 Number of hydrophones at receiver: 1
 Maximum water depth: 100.0 m
 Nominal frequency: 5000.0 Hz
 Nominal bandwidth: 1000.0 Hz

Propagation modeling

Nominal transmission loss: 70.3 dB
 50% confidence interval: [69.8, 73.1] dB
 90% confidence interval: [65.4, 74.5] dB

Nominal 6 dB delay spread: 294.8 ms
 50% confidence interval: [299.3, 291.2] ms

90% confidence interval: [307.4, 280.5] ms

Communication performance

Communication scheme: FHBFSK2 (see LRComms D3)
 Frame length: 168 user bits (+ 8 bit CRC)
 Frame duration: 10.0 seconds
 User data rate: 16.8 bps
 Frame success rate: 100.0 %
 Diversity usage: 0.0 %
 Uncoded BER (min/median/max): 0.0 / 0.0 / 0.0

Data transfer recommendations

Datagram size: 256 bytes
 Minimum number of frames: 13

# of frames transmitted	Probability of successful transfer	
	No diversity	With diversity
13	100.0%	100.0%

15 km, SSP1, MUD

Simulation summary

Nominal transmission range: 15000.0 500.0 m
 Nominal transmitter depth: 5.0 2.0 m
 Nominal receiver depth: 10.0 5.0 m
 Number of hydrophones at receiver: 1
 Maximum water depth: 100.0 m
 Nominal frequency: 5000.0 Hz
 Nominal bandwidth: 1000.0 Hz

Propagation modeling

Nominal transmission loss: 92.6 dB
 50% confidence interval: [90.3, 94.7] dB
 90% confidence interval: [89.8, 99.9] dB

Nominal 6 dB delay spread: 0.0 ms
 50% confidence interval: [0.0, 0.0] ms
 90% confidence interval: [0.0, 0.0] ms

Communication performance

Communication scheme: FHBFSK2 (see LRComms D3)
 Frame length: 168 user bits (+ 8 bit CRC)
 Frame duration: 10.0 seconds
 User data rate: 16.8 bps
 Frame success rate: 100.0 %
 Diversity usage: 0.0 %
 Uncoded BER (min/median/max): 0.0 / 0.0 / 0.0222

Data transfer recommendations

Datagram size: 256 bytes
 Minimum number of frames: 13

# of frames transmitted	Probability of successful transfer	
	No diversity	With diversity
13	100.0%	100.0%

15 km, SSP1, SAND

Simulation summary

Nominal transmission range: 15000.0 500.0 m
 Nominal transmitter depth: 5.0 2.0 m
 Nominal receiver depth: 10.0 5.0 m
 Number of hydrophones at receiver: 1
 Maximum water depth: 100.0 m
 Nominal frequency: 5000.0 Hz
 Nominal bandwidth: 1000.0 Hz

Propagation modeling

Nominal transmission loss: 70.2 dB
 50% confidence interval: [68.3, 74.4] dB
 90% confidence interval: [64.8, 80.2] dB

Nominal 6 dB delay spread: 221.8 ms
 50% confidence interval: [229.1, 214.2] ms
 90% confidence interval: [232.0, 212.1] ms

Communication performance

Communication scheme: FHBFSK2 (see LRComms D3)
 Frame length: 168 user bits (+ 8 bit CRC)
 Frame duration: 10.0 seconds
 User data rate: 16.8 bps
 Frame success rate: 100.0 %
 Diversity usage: 0.0 %
 Uncoded BER (min/median/max): 0.0 / 0.0 / 0.0

Data transfer recommendations

Datagram size: 256 bytes
 Minimum number of frames: 13

# of frames transmitted	Probability of successful transfer	
	No diversity	With diversity
13	100.0%	100.0%

27 km, ISO, MUD

Simulation summary

Nominal transmission range: 27000.0 500.0 m
 Nominal transmitter depth: 5.0 2.0 m
 Nominal receiver depth: 10.0 5.0 m
 Number of hydrophones at receiver: 1
 Maximum water depth: 100.0 m
 Nominal frequency: 5000.0 Hz
 Nominal bandwidth: 1000.0 Hz

Propagation modeling

Nominal transmission loss: 89.4 dB
 50% confidence interval: [86.2, 92.7] dB
 90% confidence interval: [85.1, 100.2] dB

Nominal 6 dB delay spread: 2.1 ms
 50% confidence interval: [2.1, 2.0] ms
 90% confidence interval: [2.1, 2.0] ms

Communication performance

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Communication scheme: FHBFSK2 (see LRComms D3)
Frame length: 168 user bits (+ 8 bit CRC)
Frame duration: 10.0 seconds
User data rate: 16.8 bps
Frame success rate: 90.0 %
Diversity usage: 0.0 %
Uncoded BER (min/median/max): 0.0 / 0.0 / 0.5302

```

Data transfer recommendations

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Datagram size: 256 bytes
Minimum number of frames: 13

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# of frames transmitted	Probability of successful transfer	
	No diversity	With diversity
13	25.4%	25.4%
14	58.5%	58.5%
15	81.6%	81.6%
16	93.2%	93.2%
17	97.8%	97.8%
18	99.4%	99.4%
19	99.8%	99.8%
20	100.0%	100.0%

27 km, ISO, SAND

Simulation summary

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Nominal transmission range: 27000.0 500.0 m
Nominal transmitter depth: 5.0 2.0 m
Nominal receiver depth: 10.0 5.0 m
Number of hydrophones at receiver: 1
Maximum water depth: 100.0 m
Nominal frequency: 5000.0 Hz
Nominal bandwidth: 1000.0 Hz

```

Propagation modeling

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Nominal transmission loss: 81.9 dB
50% confidence interval: [79.4, 86.8] dB
90% confidence interval: [76.6, 87.5] dB

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Nominal 6 dB delay spread: 251.8 ms
 50% confidence interval: [254.2, 248.6] ms
 90% confidence interval: [257.2, 246.4] ms

Communication performance

Communication scheme: FHBFSK2 (see LRComms D3)
 Frame length: 168 user bits (+ 8 bit CRC)
 Frame duration: 10.0 seconds
 User data rate: 16.8 bps
 Frame success rate: 100.0 %
 Diversity usage: 0.0 %
 Uncoded BER (min/median/max): 0.0 / 0.0 / 0.0

Data transfer recommendations

Datagram size: 256 bytes
 Minimum number of frames: 13

# of frames	Probability of successful transfer	
transmitted	No diversity	With diversity
13	100.0%	100.0%

27 km, SSP1, MUD

Simulation summary

Nominal transmission range: 27000.0 500.0 m
 Nominal transmitter depth: 5.0 2.0 m
 Nominal receiver depth: 10.0 5.0 m
 Number of hydrophones at receiver: 1
 Maximum water depth: 100.0 m
 Nominal frequency: 5000.0 Hz
 Nominal bandwidth: 1000.0 Hz

Propagation modeling

Nominal transmission loss: 109.4 dB
 50% confidence interval: [105.0, 119.4] dB
 90% confidence interval: [103.6, 121.6] dB

Nominal 6 dB delay spread: 0.0 ms
 50% confidence interval: [0.0, 0.0] ms
 90% confidence interval: [0.0, 0.0] ms

Communication performance

Communication scheme: FHBFSK2 (see LRComms D3)
 Frame length: 168 user bits (+ 8 bit CRC)
 Frame duration: 10.0 seconds
 User data rate: 16.8 bps
 Frame success rate: 30.0 %
 Diversity usage: 0.0 %
 Uncoded BER (min/median/max): 0.0363 / 0.3095 / 0.5161

Data transfer recommendations

Datagram size: 256 bytes
 Minimum number of frames: 13

# of frames transmitted	Probability of successful transfer	
	No diversity	With diversity
13	0.0%	0.0%
14	0.0%	0.0%
15	0.0%	0.0%
16	0.0%	0.0%
17	0.0%	0.0%
18	0.0%	0.0%
19	0.1%	0.1%
20	0.1%	0.1%
21	0.2%	0.2%
22	0.4%	0.4%
23	0.7%	0.7%
24	1.2%	1.2%
25	1.7%	1.7%
26	2.6%	2.6%

27 km, SSP1, SAND

Simulation summary

Nominal transmission range: 27000.0 500.0 m
 Nominal transmitter depth: 5.0 2.0 m
 Nominal receiver depth: 10.0 5.0 m
 Number of hydrophones at receiver: 1

Maximum water depth: 100.0 m
 Nominal frequency: 5000.0 Hz
 Nominal bandwidth: 1000.0 Hz

Propagation modeling

Nominal transmission loss: 82.2 dB
 50% confidence interval: [80.8, 85.5] dB
 90% confidence interval: [76.3, 89.1] dB

Nominal 6 dB delay spread: 226.7 ms
 50% confidence interval: [237.6, 222.1] ms
 90% confidence interval: [238.4, 206.2] ms

Communication performance

Communication scheme: FHBFSK2 (see LRComms D3)
 Frame length: 168 user bits (+ 8 bit CRC)
 Frame duration: 10.0 seconds
 User data rate: 16.8 bps
 Frame success rate: 100.0 %
 Diversity usage: 0.0 %
 Uncoded BER (min/median/max): 0.0 / 0.0 / 0.0

Data transfer recommendations

Datagram size: 256 bytes
 Minimum number of frames: 13

# of frames transmitted	Probability of successful transfer	
	No diversity	With diversity
13	100.0%	100.0%

45 km (24-element vertical array), ISO, MUD

Simulation summary

Nominal transmission range: 45000.0 500.0 m
 Nominal transmitter depth: 5.0 2.0 m
 Nominal receiver depth: 10.0 5.0 m
 Number of hydrophones at receiver: 24
 Maximum water depth: 100.0 m

Nominal frequency: 5000.0 Hz
 Nominal bandwidth: 1000.0 Hz

Propagation modeling

Nominal transmission loss: 102.6 dB
 50% confidence interval: [100.5, 107.8] dB
 90% confidence interval: [95.9, 112.0] dB

Nominal 6 dB delay spread: 2.4 ms
 50% confidence interval: [2.6, 2.5] ms
 90% confidence interval: [2.7, 1.2] ms

Communication performance

Communication scheme: FHBFSK2 (see LRComms D3)
 Frame length: 168 user bits (+ 8 bit CRC)
 Frame duration: 10.0 seconds
 User data rate: 16.8 bps
 Frame success rate: 99.0 %
 Diversity usage: 19.0 %
 Uncoded BER (min/median/max): 0.0 / 0.0121 / 0.1492

Data transfer recommendations

Datagram size: 256 bytes
 Minimum number of frames: 13

# of frames transmitted	Probability of successful transfer	
	No diversity	With diversity
13	5.5%	87.8%
14	19.8%	99.2%
15	39.8%	100.0%
16	59.8%	100.0%
17	75.8%	100.0%
18	86.7%	100.0%
19	93.2%	100.0%
20	96.8%	100.0%
21	98.6%	100.0%
22	99.4%	100.0%
23	99.7%	100.0%
24	99.9%	100.0%
25	100.0%	100.0%

45 km (24-element vertical array), ISO, SAND

Simulation summary

 Nominal transmission range: 45000.0 500.0 m
 Nominal transmitter depth: 5.0 2.0 m
 Nominal receiver depth: 10.0 5.0 m
 Number of hydrophones at receiver: 24
 Maximum water depth: 100.0 m
 Nominal frequency: 5000.0 Hz
 Nominal bandwidth: 1000.0 Hz

Propagation modeling

 Nominal transmission loss: 94.0 dB
 50% confidence interval: [92.9, 98.3] dB
 90% confidence interval: [87.0, 99.7] dB

 Nominal 6 dB delay spread: 216.6 ms
 50% confidence interval: [227.5, 207.3] ms
 90% confidence interval: [246.2, 176.2] ms

Communication performance

 Communication scheme: FHBFSK2 (see LRComms D3)
 Frame length: 168 user bits (+ 8 bit CRC)
 Frame duration: 10.0 seconds
 User data rate: 16.8 bps
 Frame success rate: 100.0 %
 Diversity usage: 0.0 %
 Uncoded BER (min/median/max): 0.0 / 0.0 / 0.0

Data transfer recommendations

 Datagram size: 256 bytes
 Minimum number of frames: 13

# of frames transmitted	Probability of successful transfer	
	No diversity	With diversity
13	100.0%	100.0%

45 km (24-element vertical array), SSP1, MUD

Simulation summary

Nominal transmission range: 45000.0 500.0 m
 Nominal transmitter depth: 5.0 2.0 m
 Nominal receiver depth: 10.0 5.0 m
 Number of hydrophones at receiver: 24
 Maximum water depth: 100.0 m
 Nominal frequency: 5000.0 Hz
 Nominal bandwidth: 1000.0 Hz

Propagation modeling

Nominal transmission loss: 112.0 dB
 50% confidence interval: [108.2, 121.5] dB
 90% confidence interval: [105.1, 130.5] dB

Nominal 6 dB delay spread: 0.0 ms
 50% confidence interval: [0.0, 0.0] ms
 90% confidence interval: [0.0, 0.0] ms

Communication performance

Communication scheme: FHBFSK2 (see LRComms D3)
 Frame length: 168 user bits (+ 8 bit CRC)
 Frame duration: 10.0 seconds
 User data rate: 16.8 bps
 Frame success rate: 70.0 %
 Diversity usage: 70.0 %
 Uncoded BER (min/median/max): 0.0 / 0.0454 / 0.5141

Data transfer recommendations

Datagram size: 256 bytes
 Minimum number of frames: 13

# of frames transmitted	Probability of successful transfer	
	No diversity	With diversity
13	0.0%	1.0%
14	0.0%	4.7%
15	0.0%	12.7%
16	0.0%	24.6%
17	0.0%	38.9%
18	0.0%	53.4%
19	0.0%	66.6%

20	0.0%	77.2%
21	0.0%	85.2%
22	0.0%	90.8%
23	0.0%	94.5%
24	0.0%	96.9%
25	0.0%	98.3%
26	0.0%	99.1%

45 km (24-element vertical array), SSP1, SAND

Simulation summary

Nominal transmission range: 45000.0 500.0 m
 Nominal transmitter depth: 5.0 2.0 m
 Nominal receiver depth: 10.0 5.0 m
 Number of hydrophones at receiver: 24
 Maximum water depth: 100.0 m
 Nominal frequency: 5000.0 Hz
 Nominal bandwidth: 1000.0 Hz

Propagation modeling

Nominal transmission loss: 93.9 dB
 50% confidence interval: [92.3, 97.4] dB
 90% confidence interval: [86.4, 102.6] dB

Nominal 6 dB delay spread: 236.9 ms
 50% confidence interval: [252.7, 223.6] ms
 90% confidence interval: [254.6, 208.7] ms

Communication performance

Communication scheme: FHBFSK2 (see LRComms D3)
 Frame length: 168 user bits (+ 8 bit CRC)
 Frame duration: 10.0 seconds
 User data rate: 16.8 bps
 Frame success rate: 100.0 %
 Diversity usage: 0.0 %
 Uncoded BER (min/median/max): 0.0 / 0.0 / 0.0

Data transfer recommendations

Datagram size: 256 bytes
 Minimum number of frames: 13

# of frames transmitted	Probability of successful transfer	
	No diversity	With diversity
13	100.0%	100.0%

45 km (24-element horizontal array), ISO, MUD

Simulation summary

Nominal transmission range: 45000.0 500.0 m
 Nominal transmitter depth: 5.0 2.0 m
 Nominal receiver depth: 10.0 5.0 m
 Number of hydrophones at receiver: 24
 Maximum water depth: 100.0 m
 Nominal frequency: 5000.0 Hz
 Nominal bandwidth: 1000.0 Hz

Propagation modeling

Nominal transmission loss: 100.5 dB
 50% confidence interval: [97.6, 105.6] dB
 90% confidence interval: [94.5, Inf] dB

Nominal 6 dB delay spread: 2.5 ms
 50% confidence interval: [2.8, 2.7] ms
 90% confidence interval: [2.9, 0.0] ms

Communication performance

Communication scheme: FHBFSK2 (see LRComms D3)
 Frame length: 168 user bits (+ 8 bit CRC)
 Frame duration: 10.0 seconds
 User data rate: 16.8 bps
 Frame success rate: 90.0 %
 Diversity usage: 16.0 %
 Uncoded BER (min/median/max): 0.0 / 0.0131 / 0.4879

Data transfer recommendations

Datagram size: 256 bytes

Minimum number of frames: 13

# of frames transmitted	Probability of successful transfer	
	No diversity	With diversity
13	2.0%	25.4%
14	8.7%	58.5%
15	21.0%	81.6%
16	37.0%	93.2%
17	53.6%	97.8%
18	68.2%	99.4%
19	79.7%	99.8%
20	87.8%	100.0%
21	93.0%	100.0%
22	96.2%	100.0%
23	98.0%	100.0%
24	99.0%	100.0%
25	99.5%	100.0%
26	99.8%	100.0%

45 km (24-element horizontal array), ISO, SAND

Simulation summary

Nominal transmission range: 45000.0 500.0 m
 Nominal transmitter depth: 5.0 2.0 m
 Nominal receiver depth: 10.0 5.0 m
 Number of hydrophones at receiver: 24
 Maximum water depth: 100.0 m
 Nominal frequency: 5000.0 Hz
 Nominal bandwidth: 1000.0 Hz

Propagation modeling

Nominal transmission loss: 86.8 dB
 50% confidence interval: [84.4, 90.8] dB
 90% confidence interval: [80.6, 100.9] dB

 Nominal 6 dB delay spread: 228.0 ms
 50% confidence interval: [229.3, 225.3] ms
 90% confidence interval: [238.4, 224.9] ms

Communication performance

Communication scheme: FHBFSK2 (see LRComms D3)
 Frame length: 168 user bits (+ 8 bit CRC)
 Frame duration: 10.0 seconds
 User data rate: 16.8 bps
 Frame success rate: 100.0 %
 Diversity usage: 0.0 %
 Uncoded BER (min/median/max): 0.0 / 0.0 / 0.0

Data transfer recommendations

Datagram size: 256 bytes
 Minimum number of frames: 13

# of frames transmitted	Probability of successful transfer	
	No diversity	With diversity
13	100.0%	100.0%

45 km (24-element horizontal array), SSP1, MUD

Simulation summary

Nominal transmission range: 45000.0 500.0 m
 Nominal transmitter depth: 5.0 2.0 m
 Nominal receiver depth: 10.0 5.0 m
 Number of hydrophones at receiver: 24
 Maximum water depth: 100.0 m
 Nominal frequency: 5000.0 Hz
 Nominal bandwidth: 1000.0 Hz

Propagation modeling

Nominal transmission loss: 118.9 dB
 50% confidence interval: [117.8, 125.1] dB
 90% confidence interval: [111.1, 130.3] dB

Nominal 6 dB delay spread: 0.0 ms
 50% confidence interval: [0.0, 0.0] ms
 90% confidence interval: [0.0, 0.0] ms

Communication performance

Communication scheme: FHBFSK2 (see LRComms D3)

Frame length: 168 user bits (+ 8 bit CRC)
 Frame duration: 10.0 seconds
 User data rate: 16.8 bps
 Frame success rate: 20.0 %
 Diversity usage: 20.0 %
 Uncoded BER (min/median/max): 0.0 / 0.4234 / 0.5242

Data transfer recommendations

Datagram size: 256 bytes
 Minimum number of frames: 13

# of frames transmitted	Probability of successful transfer	
	No diversity	With diversity
13	0.0%	0.0%
14	0.0%	0.0%
15	0.0%	0.0%
16	0.0%	0.0%
17	0.0%	0.0%
18	0.0%	0.0%
19	0.0%	0.0%
20	0.0%	0.0%
21	0.0%	0.0%
22	0.0%	0.0%
23	0.0%	0.0%
24	0.0%	0.0%
25	0.0%	0.0%
26	0.0%	0.1%

45 km (24-element horizontal array), SSP1, SAND

Simulation summary

Nominal transmission range: 45000.0 500.0 m
 Nominal transmitter depth: 5.0 2.0 m
 Nominal receiver depth: 10.0 5.0 m
 Number of hydrophones at receiver: 24
 Maximum water depth: 100.0 m
 Nominal frequency: 5000.0 Hz
 Nominal bandwidth: 1000.0 Hz

Propagation modeling

Nominal transmission loss: 90.2 dB
 50% confidence interval: [87.9, 94.9] dB
 90% confidence interval: [83.3, 106.1] dB

Nominal 6 dB delay spread: 254.8 ms
 50% confidence interval: [256.1, 253.4] ms
 90% confidence interval: [256.9, 252.6] ms

Communication performance

Communication scheme: FHBFSK2 (see LRComms D3)
 Frame length: 168 user bits (+ 8 bit CRC)
 Frame duration: 10.0 seconds
 User data rate: 16.8 bps
 Frame success rate: 100.0 %
 Diversity usage: 10.0 %
 Uncoded BER (min/median/max): 0.0 / 0.0 / 0.0847

Data transfer recommendations

Datagram size: 256 bytes
 Minimum number of frames: 13

# of frames transmitted	Probability of successful transfer	
	No diversity	With diversity
13	25.4%	100.0%
14	58.5%	100.0%
15	81.6%	100.0%
16	93.2%	100.0%
17	97.8%	100.0%
18	99.4%	100.0%
19	99.8%	100.0%
20	100.0%	100.0%

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